Assignment-19.1

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Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

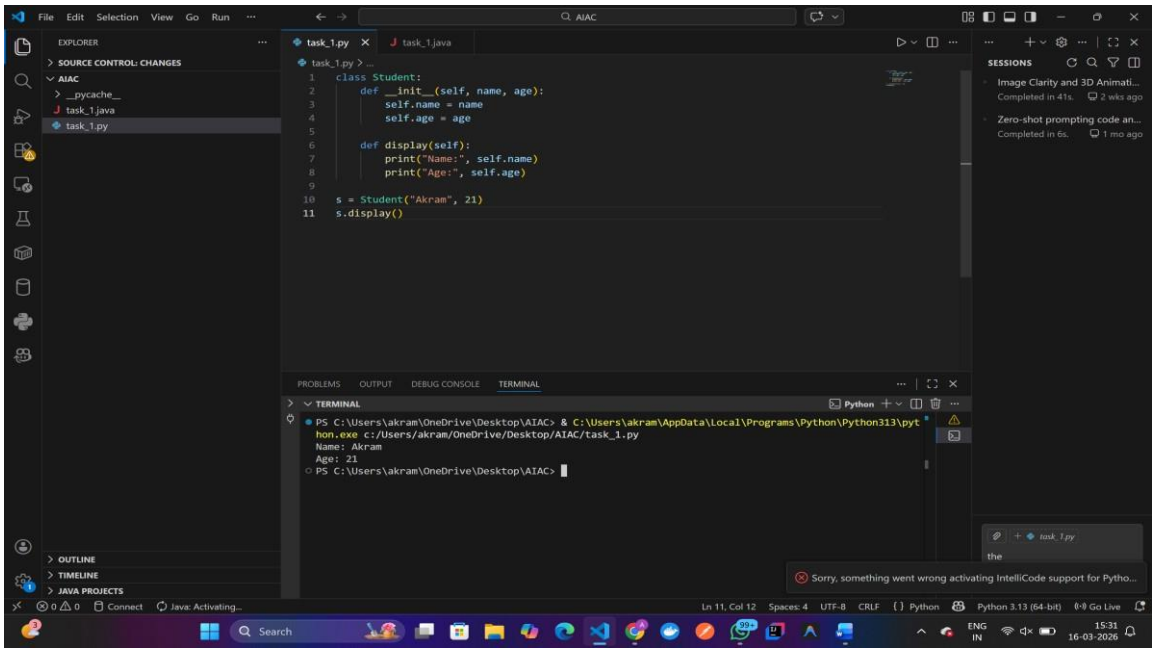
Instructions:

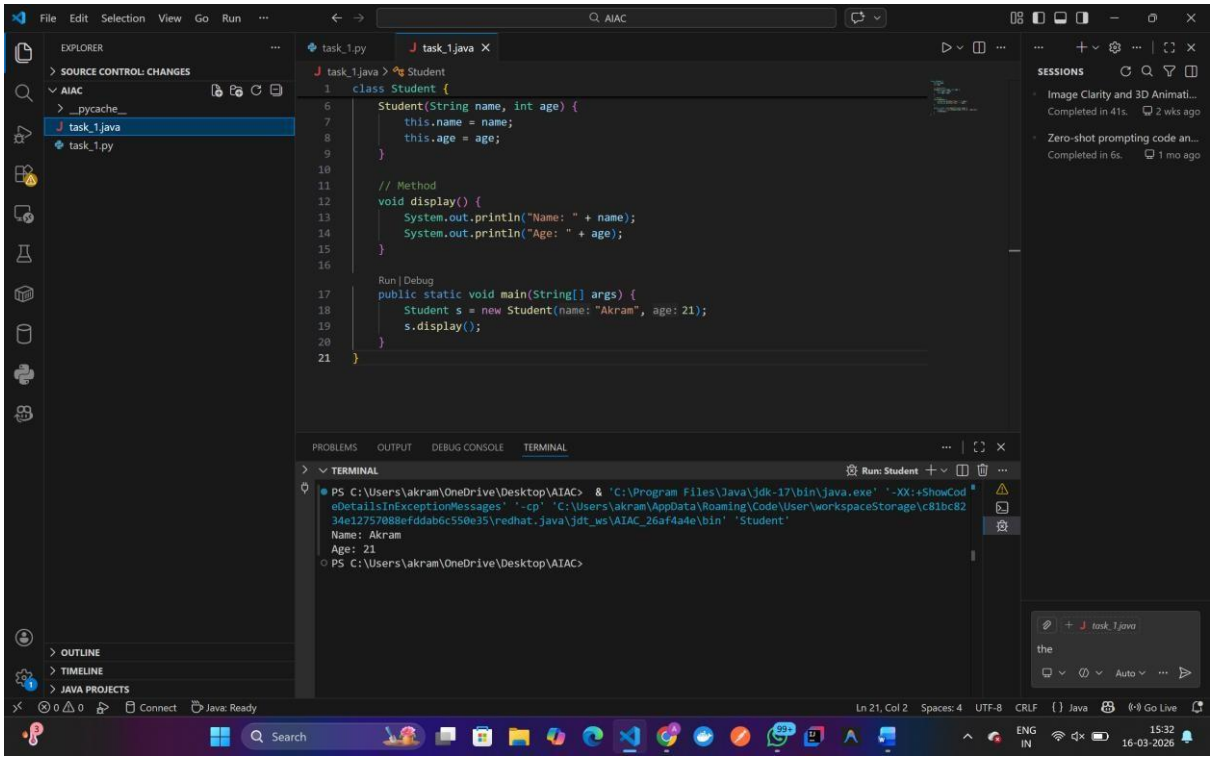
- Prompt AI to convert a Python class with attributes and methods into equivalent Java code.
- Verify that constructors, method signatures, and data types are properly handled.
- Run both versions (Python & Java) to compare expected outputs.

Expected Output:

- Equivalent working Java class translated from Python.

CODE:





Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

• Ask AI to translate a recursive C++ factorial function into

Python.

• Validate correctness by testing with sample inputs.

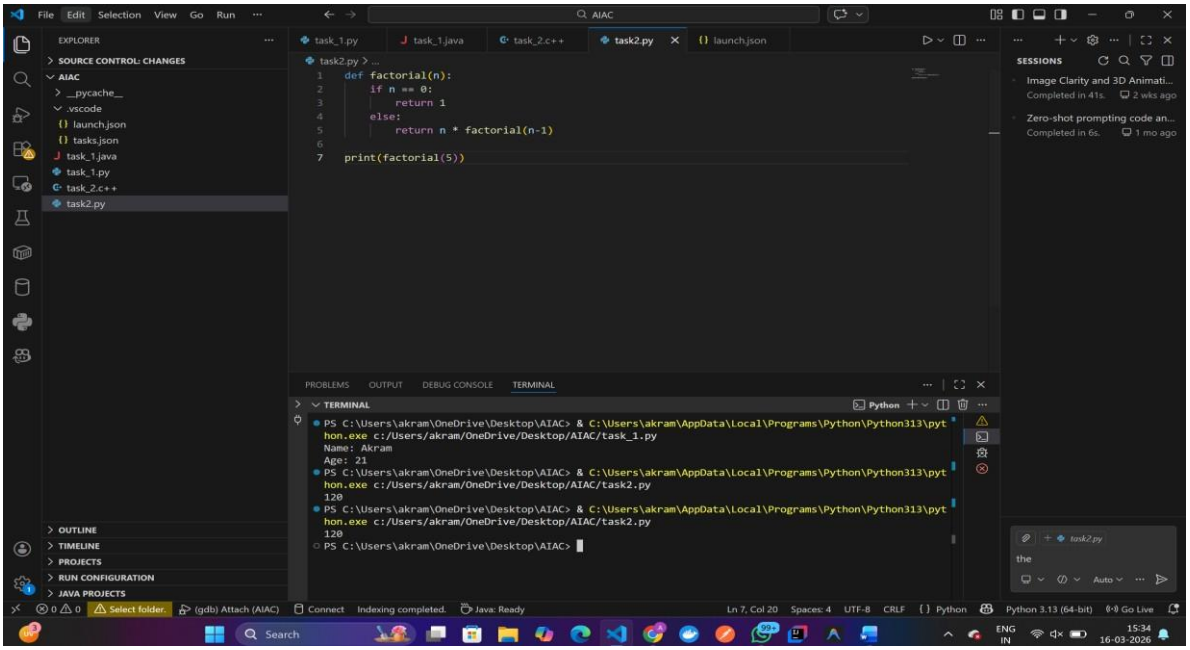
• Ensure Python version follows Pythonic conventions.

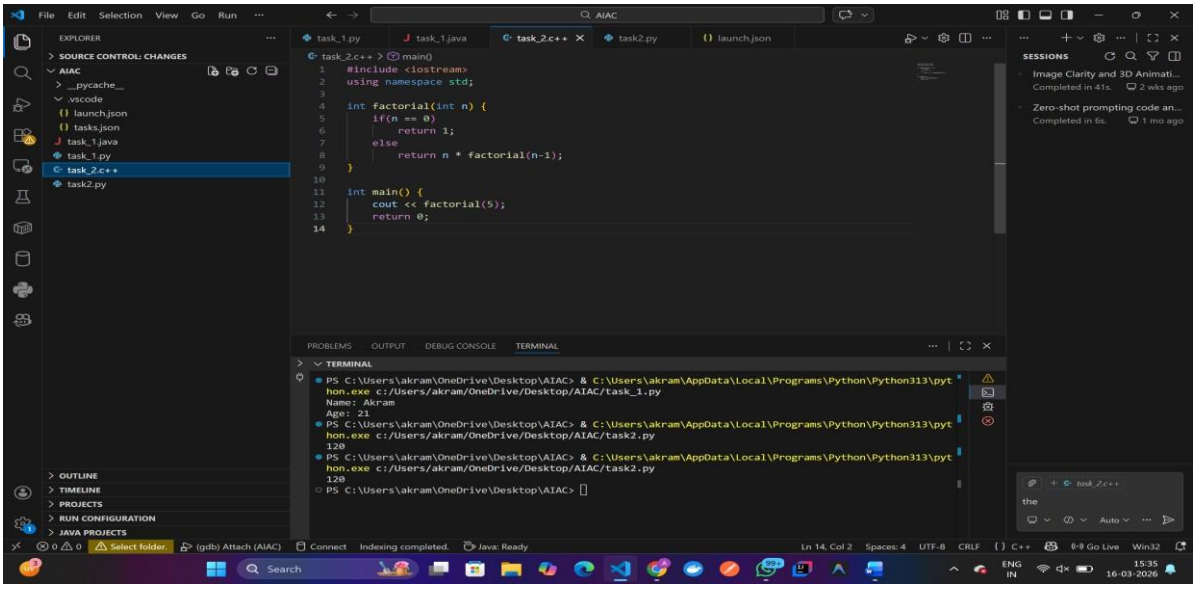
Expected Output:

• A correct Python implementation equivalent to the original

C++ factorial function.

CODE:





Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in

Python.

Instructions:

• Provide AI with a SQL query (e.g., SELECT name, salary

FROM employees WHERE salary > 50000).

• Ask AI to generate equivalent Pandas code.

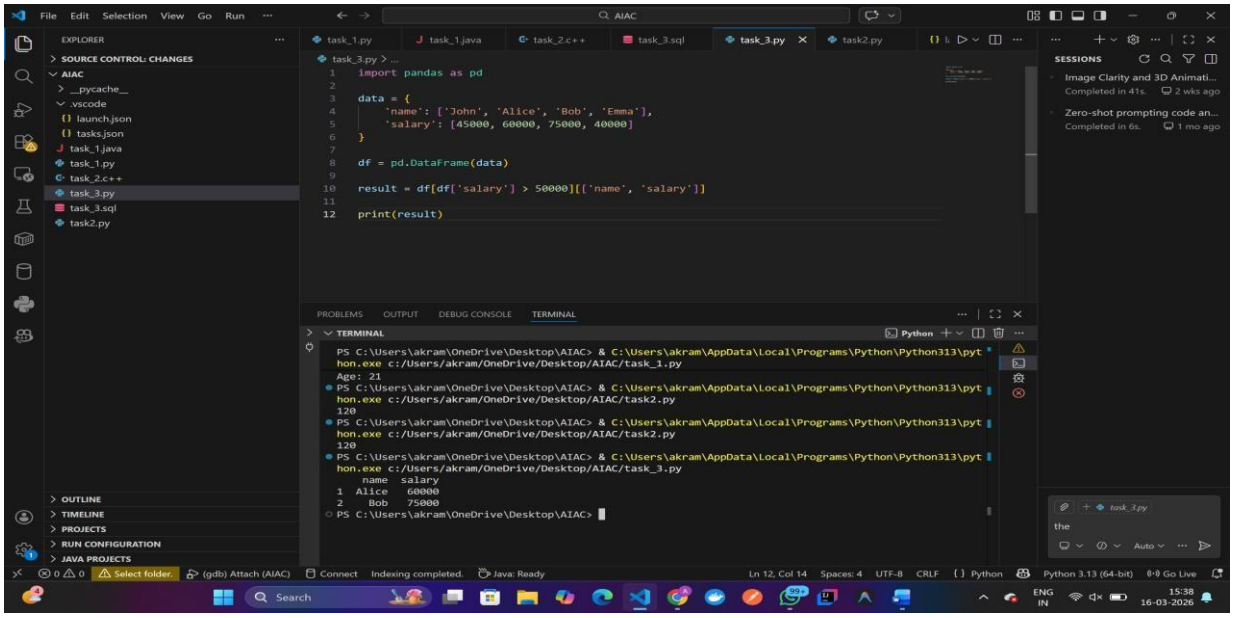
• Test the generated code on a sample DataFrame.

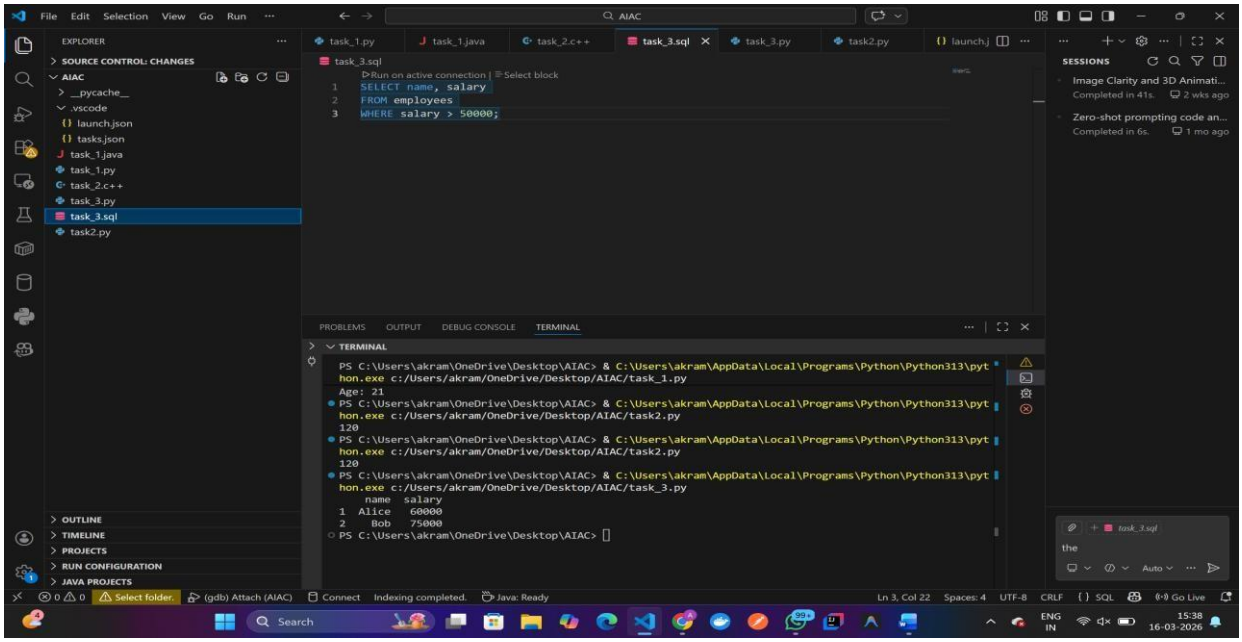
Expected Output:

• Equivalent Pandas query that retrieves the correct subset of

data.

CODE:





Task 4 – Real-Time Application: Multi-Language Snippet

Converter

Scenario:

Students will choose a real-time use case where they need code in multiple

languages. Example:

• Converting a Python web scraping snippet into JavaScript for browser automation.

• Translating a Java method into C# for enterprise applications.

Instructions:

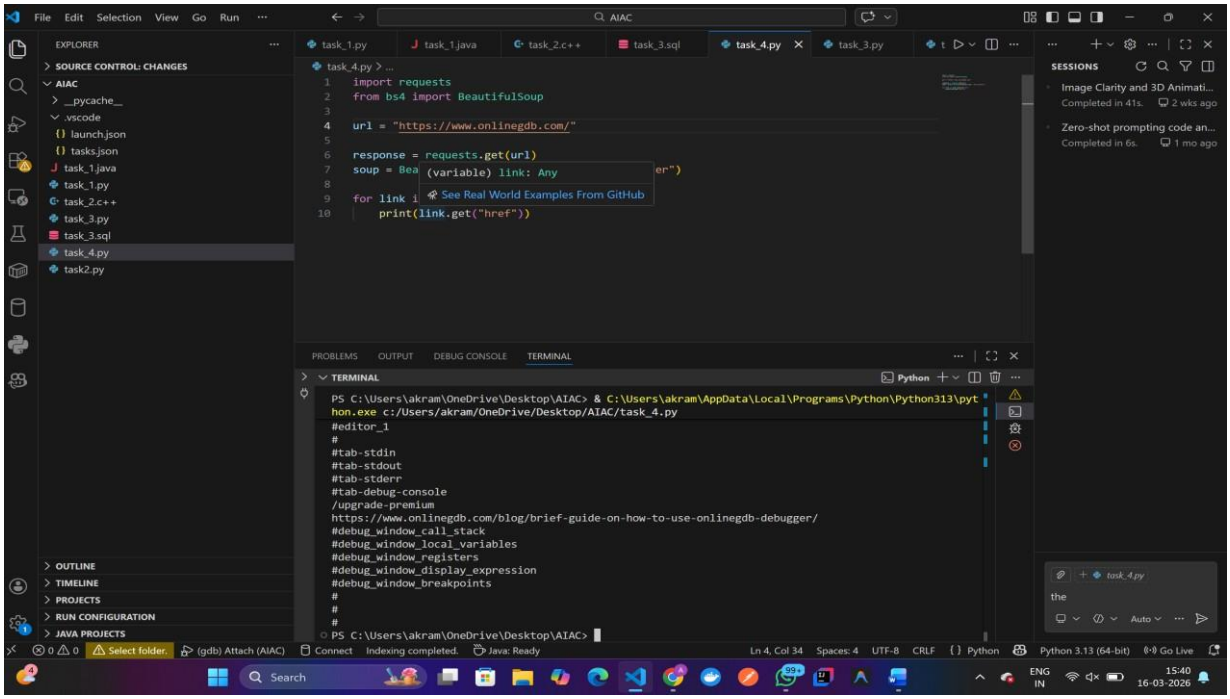
• Prompt AI to translate code between two chosen languages.

• Test both implementations to ensure they produce the same results.

• Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

• Working equivalent code snippets in at least two different programming languages CODE:



File Edit Selection View Go Run ... AIAC

EXPLORER

SOURCE CONTROL: CHANGES

AIAC
- __pycache__
- .vscode
  - launch.json
  - tasks.json
- task_1.java
- task_1.py
- task_2.c++
- task_3.py
- task_3.sql
- task_4.js
- task_4.py
- task2.py

OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS

task_1.py | task_1.java | task_2.c++ | task_3.sql | task_4.py | task_4.js | task_3.p

task_4.js > then() callback

```javascript
fetch("https://www.onlinegdb.com/")
  .then(response => response.text())
  .then(data => {
    const parser = new DOMParser();
    const doc = parser.parseFromString(data, "text/html");

    const links = doc.querySelectorAll("a");
    links.forEach(link => {
      console.log(link.href);
    });
  });
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

TERMINAL

```
PS C:\Users\akram\OneDrive\Desktop\AIAC> & C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/AIAC/task_4.py
#editor_1
#
#tab-stdin
#tab-stdout
#tab-stderr
#tab-debug-console
/upgrade-premium
https://www.onlinegdb.com/blog/brief-guide-on-how-to-use-onlinegdb-debugger/
#debug_window_call_stack
#debug_window_local_variables
#debug_window_registers
#debug_window_display_expression
#debug_window_breakpoints
#
#
#
PS C:\Users\akram\OneDrive\Desktop\AIAC>
```

SESSIONS
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- Zero-shot prompting code an... Completed in 6s. 1 mo ago

task_4.js
the
Auto

0 0 Select folder. (gdb) Attach (AIAC) Connect Indexing completed. Java: Ready Ln 7, Col 45 Spaces: 4 UTF-8 CRLF JavaScript Go Live

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